

GEOGRAPHY

PHYSICAL GEOGRAPHY

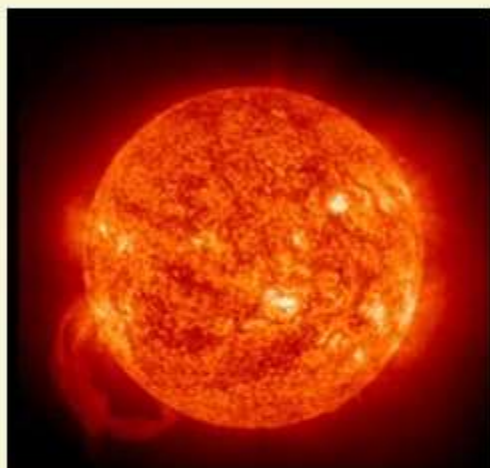


MODULE -5

- HOW SUN PROVIDES HEAT AND LIGHT?
- DOES THE SUN ROTATE?
- SUNSPOTS AND MAGNETIC FIELD
- SOLAR FLARES AND CMEs



Why the Sun has got such high luminosity?

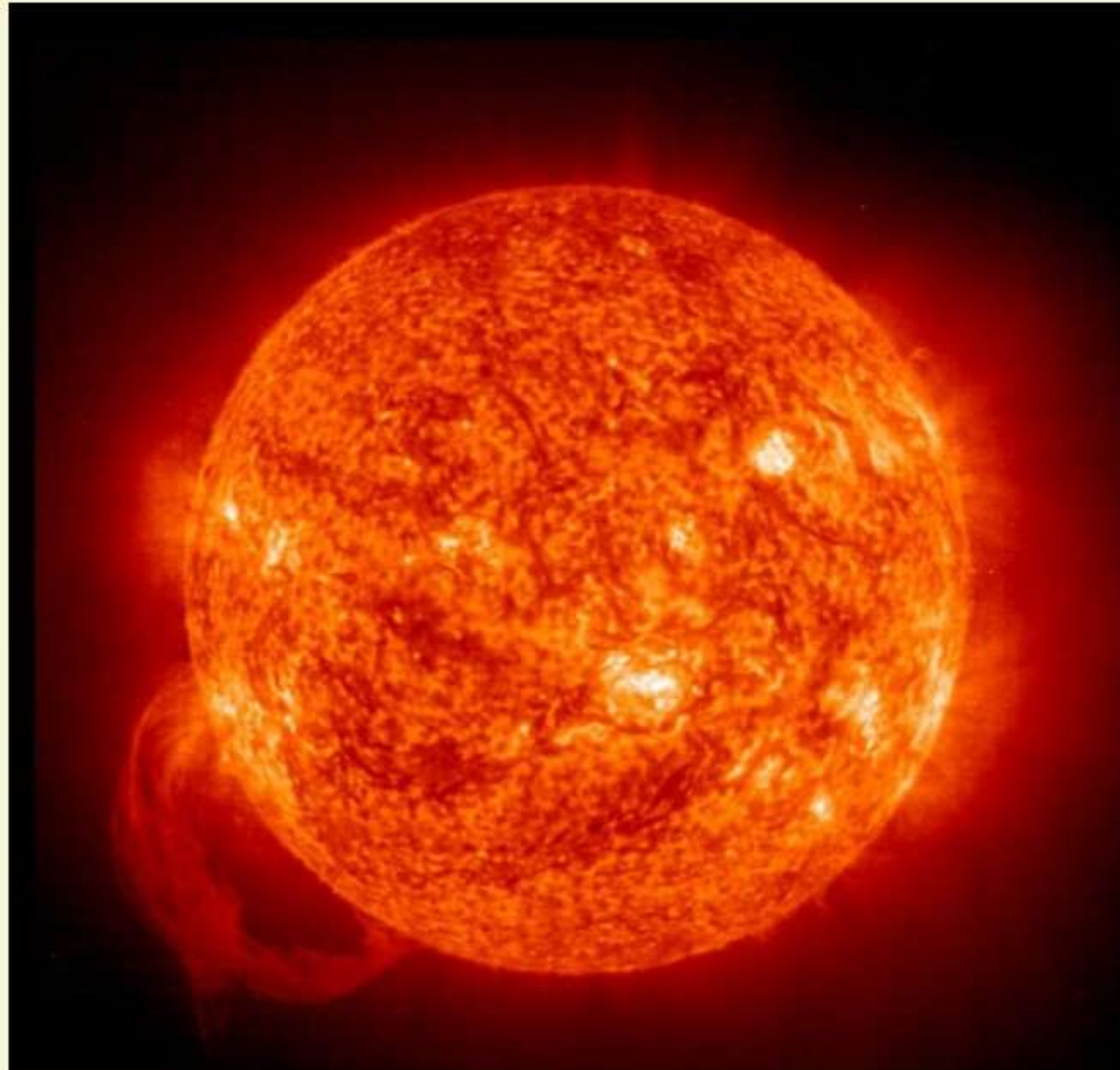


SOLAR CYCLE

- Solar cycle or solar magnetic activity cycle is the nearly periodic 11-year change in the Sun's activity.
- These effect sunspots, flares etc.

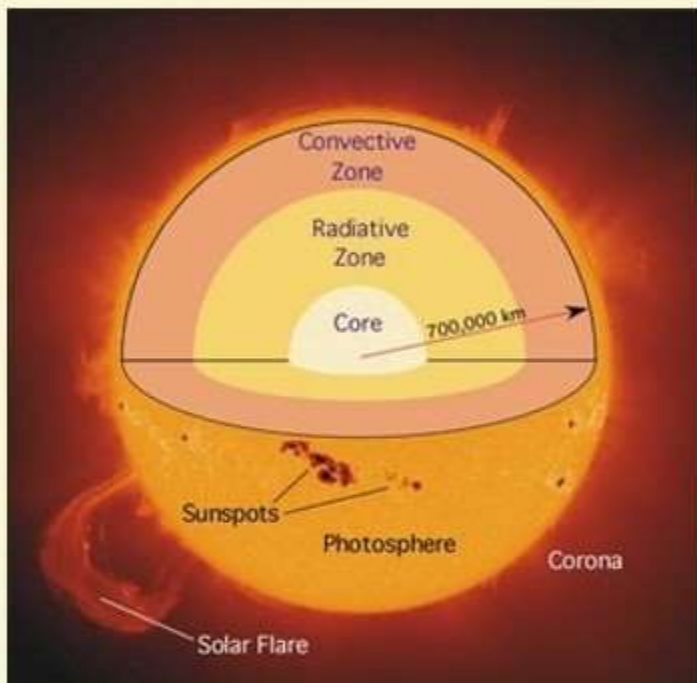
- The luminosity is 4×10^{26} MW.
- This luminosity is due to the result of Hydrogen to Helium fusion reactions that occur in the central regions.
- The energy output from the Sun is not constant but, varies with the solar cycle.
- The hottest and most active regions appear white.
- Darker areas indicate cooler temperatures.

Sun is a big ball of hot plasma (ionised gas) that holds itself by fusing hydrogen nuclei into helium nuclei in its central regions.





How does the Sun provide heat and light?



The Sun has three main parts:

- *The core.*
- *The radiative zone surrounding the core.*
- *The convective zone around that.*

- The gravity of the sun pulls its mass inward, creating a lot of pressure.
- This **pressure forces atoms of hydrogen to fuse together in nuclear reactions**, like in nuclear bombs.
- The energy gives off **different kinds of light**, like infrared light, ultraviolet light and visible light.
- Microwaves, radio waves and X-rays also come from the Sun.
- The Sun **also gives off energized particles** that make up solar wind.
- The energy from solar wind warms the Earth and influences the weather.



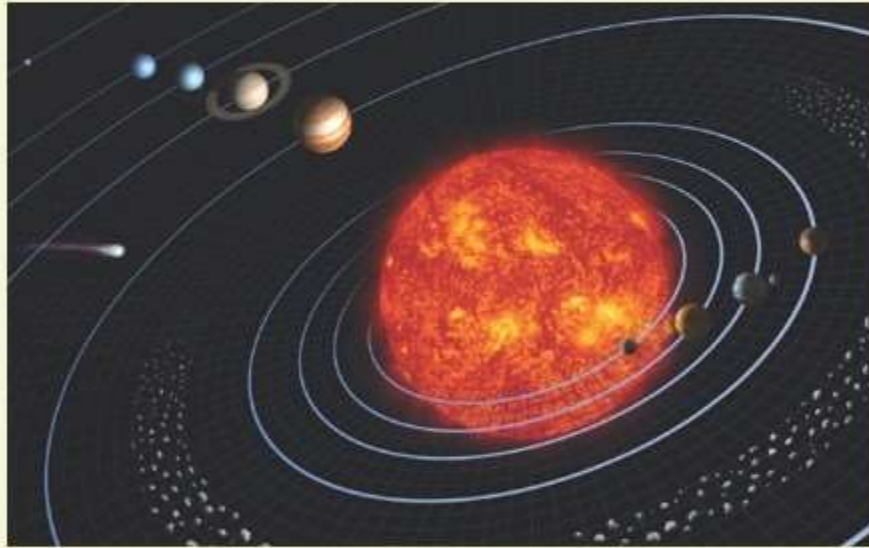
How the light comes out of the Sun?



- The light from the Sun what we see is made deep inside the Sun's interior.
- It then bounces around inside the Sun and it gradually makes its way out.
- We call the surface of the Sun the mean distance from the Sun at which the light bounces for the last time before coming out.



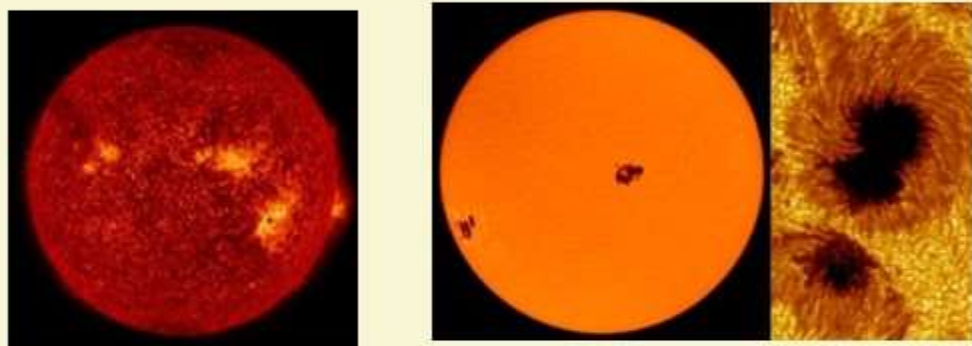
Does the Sun rotate?



- Sun rotates and it is quite different from that of the most of the planets.
- Galileo noted this nearly 400 years ago, when he observed sunspots.
- The apparent motion of the sunspots can be used to determine the rotation speed of the Sun.
- All of us know, Earth has a single rotation. Since the Sun is made up of gases, different parts rotate at different speeds.
- Sun's Equator completes one rotation in around 24 Earth days, at the Poles, it is 30 days. This is called differential rotation.



How can you say that Sunspots are associated with the magnetic field?



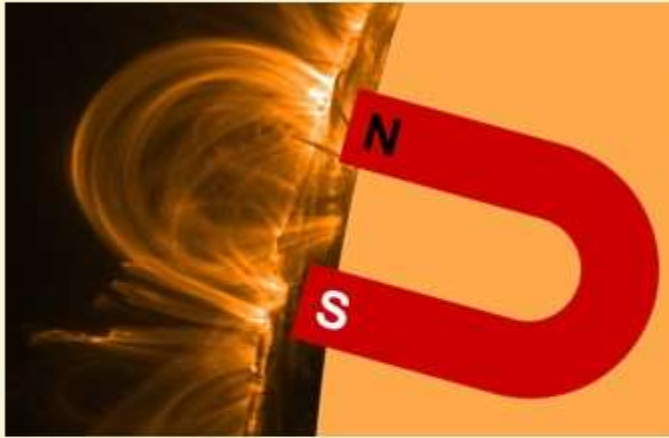
The magnetic field is caused by movements of gases in the sun's interior.

The sunspot numbers vary about the 11-year cycle.

- The dark spot which we view is a sunspot.
- They are darker than the rest of the surface, because they are cooler.
- These are the places, where the Sun's magnetic field rises up from below the Sun's surface.
- These are formed continuously as the Sun's magnetic field actively moves through the Sun.
- The centre of the sunspot has a temperature of around 3,700 kelvin, while the surrounding Photosphere has a temperature of 5,800 kelvin.



How can you say that Sunspots are associated with the magnetic field? (... contd)

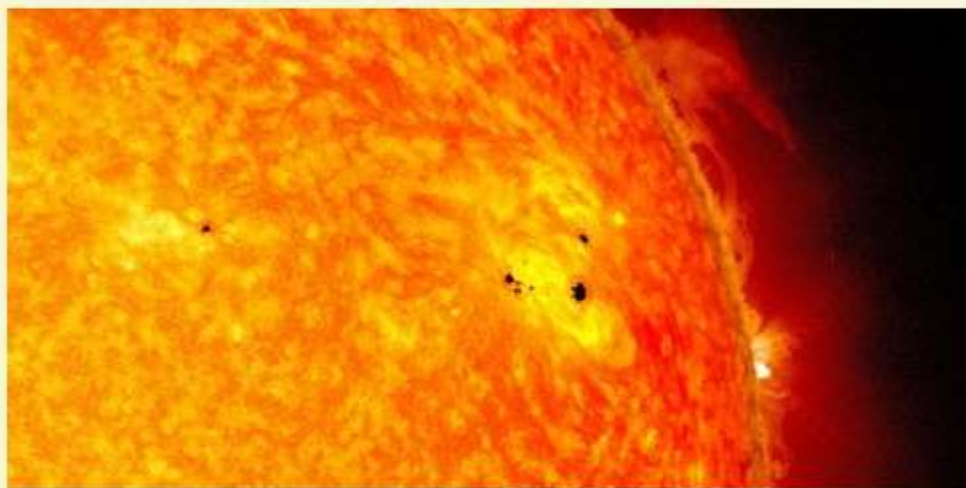


Sunspots usually appear in pairs of opposite magnetic polarity, one north pole and the other south pole. They are expending less energy and have a lower temperature.

- The reduced surface temperature is because of concentrations of magnetic field flux that inhibit convection.
- The gaseous sphere of the Sun rotates more quickly at its equator than at its poles.
- The overall magnetic field becomes distorted and twisted overtime.
- The twisted field lines eventually come out through the photosphere showing their presence as sunspots.



How do sunspots effect the Earth?



- Sunspots directly do not effect us.
- Sun's brightness may change slightly if there are many sunspots.
- Then, the Sun would be less bright and therefore less sunlight would reach the Earth.
- But, it is very very tiny so we don't recognise it.

However, related magnetic phenomena on the Sun to sunspots include solar flares, coronal mass ejections etc. These are solar "storms", a kind of "burp" where the Sun ejects high-speed and energetic particles, which are caused by the Sun's twisting and turning magnetic field.



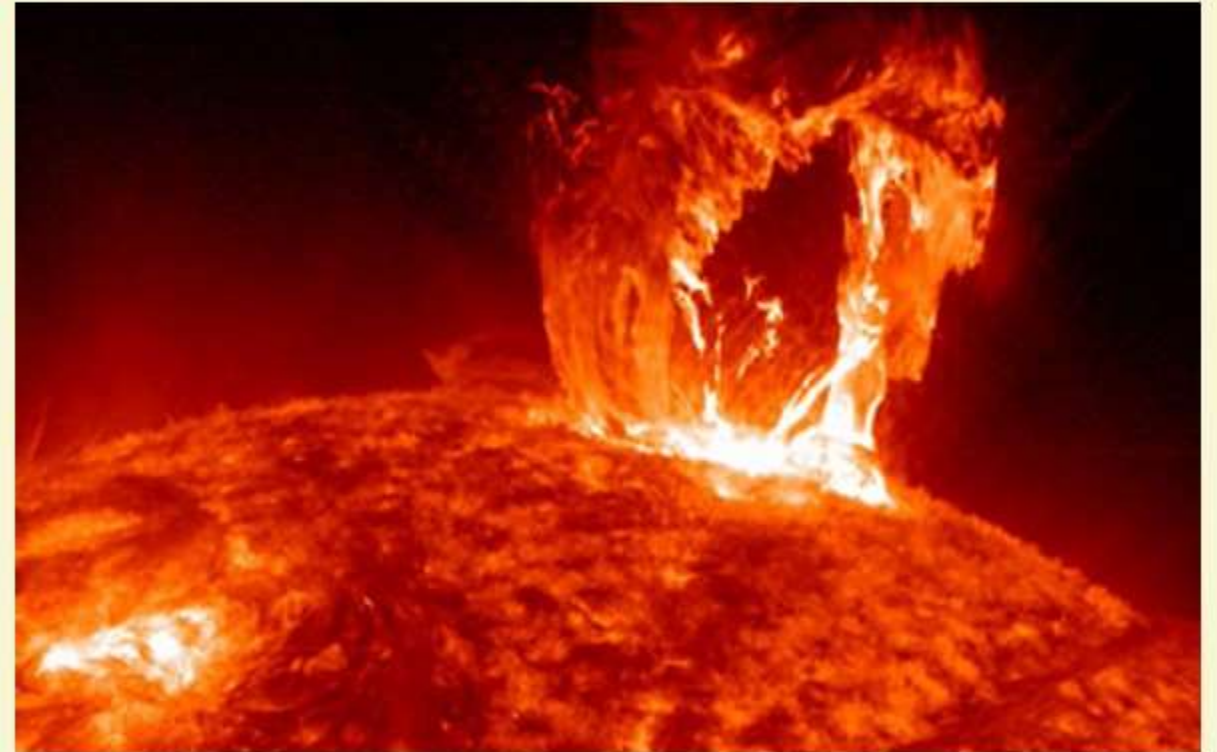
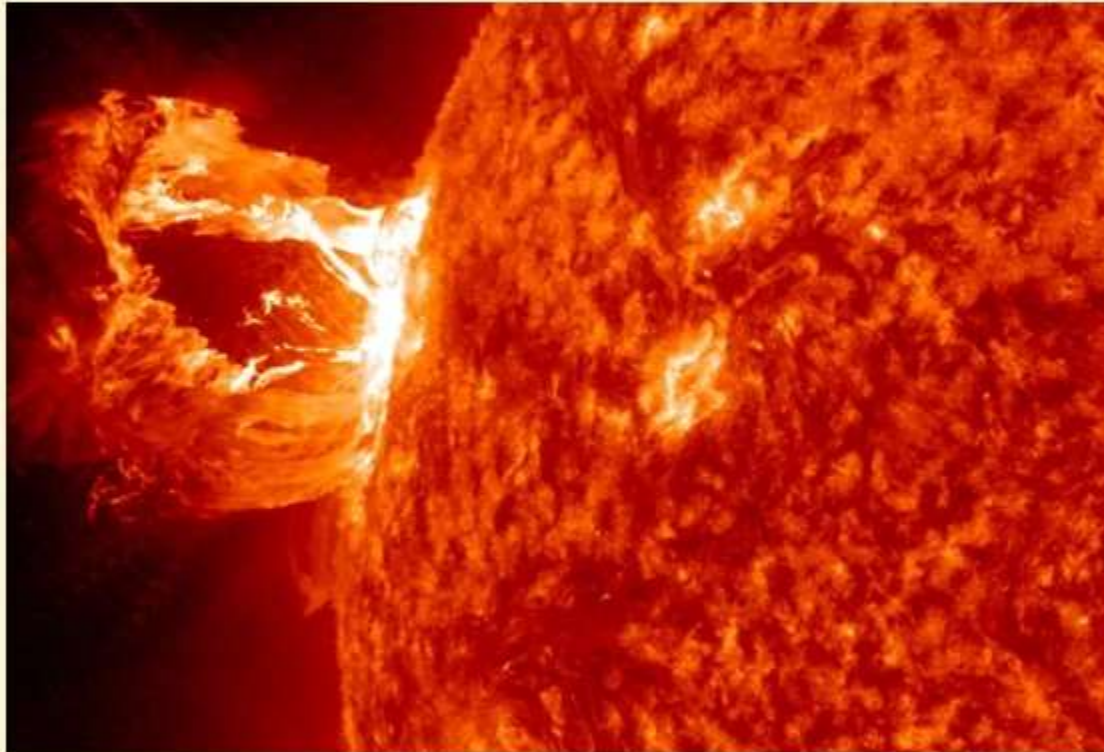
How the solar flares are formed?

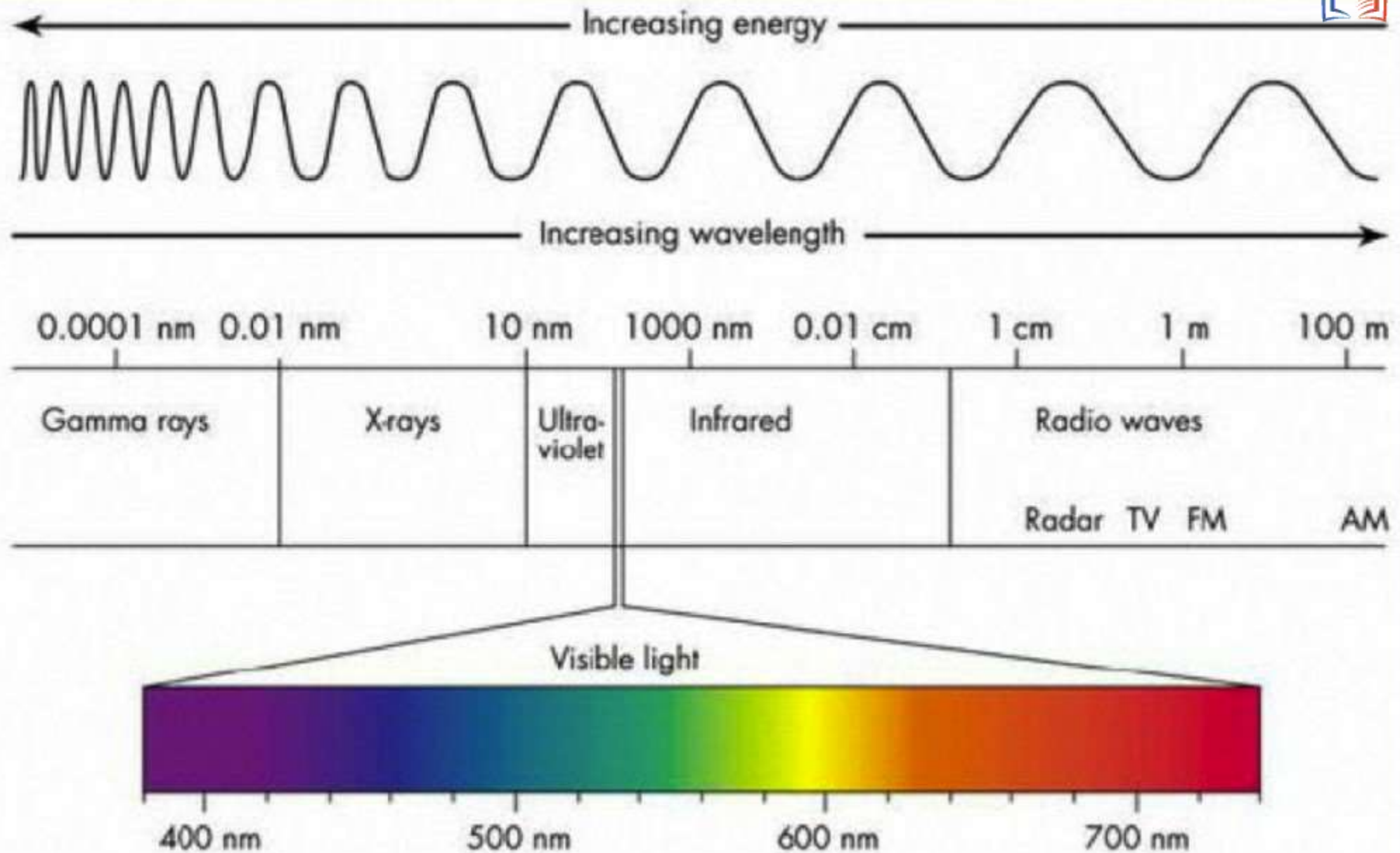


They are more active during the solar maximum part of the 11-year cycle of solar activity.

It is an active area of research.

- They usually erupt from sunspots, which are high magnetic fields.
- They seem to be related to sudden release of stored magnetic energy.
- As the solar flares push through the corona, they heat its gas to anywhere from 10 to 20 million K, occasionally reaching to 100 million K.
- They generate a burst of radiation across a wide range of electromagnetic spectrum.
- They arrive in only 8 mins, just like light.







Differentiate between solar flares and CMEs?

SOLAR FLARES

- 1 They are burst of radiation across a wide range of electromagnetic spectrum.
- 2 They travel at the speed of light in all the directions
- 3 Flares take only 8 mins.
- 4 Flares are splashes of light on the Sun.

CMEs

- 1 It is Coronal Mass Ejection.
- 2 They are giant clouds of particles hurled out into space i.e. the cloud of hot plasma and charged particles are ejected.
- 3 CMEs take nearly one to three days to reach the Earth.
- 4 CMEs are eruptions spreading out into the space.



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